

CustomerIQ

Official Product Requirements Document

Predictive Intelligence Phase — Days 5–8

Predictive Modeling → Model Comparison → ML Validation → Customer Risk Scoring

Purpose: Maintain a formal, traceable record of the work completed during CustomerIQ Days 5–8.

Current phase outcome: CustomerIQ progressed from an interpretable churn baseline to model comparison, multi-dimensional ML validation, and customer-level risk scoring.

1. Project Context & Shared Data Contract

Days 5–8 use the customer-level dataset produced from the earlier CustomerIQ data and churn-methodology work.

Component	Verified specification
Eligible customer population	3,705 customers
Target	observed_churn
Target meaning	1 = customer did not return; 0 = customer returned
Validation window	2011-09-10 to 2011-12-09 (90 days)
Predictors	Frequency; Monetary; Recency; Customer Lifespan; Average Purchase Interval; Median Purchase Interval
Feature timing	All behavioral predictors derived strictly from transactions before the cutoff
Train/test split	80/20 stratified split
Random state	42
Leakage exclusions	Customer ID; returned_in_validation; observed_churn

Important: observed_churn is an observed/provisional outcome based on the defined 90-day post-cutoff validation window. It should not be presented as a universal definition of churn.

2. DAY 5 — Predictive Modeling

Objective

Build an interpretable machine-learning baseline capable of estimating customer churn risk from pre-cutoff behavioral information.

Methodology

- Model: Logistic Regression.
- Preprocessing: StandardScaler inside a scikit-learn Pipeline.
- Predictors: six behavioral features listed in the shared data contract.
- Split: 3,705 customers into 80% training and 20% test data using stratification and random_state=42.
- Leakage control: Customer ID and future/outcome fields excluded from X.

Verified performance

Metric	Result
Train / Test	2,964 / 741
Accuracy	0.7045
Precision	0.7068
Recall	0.6272
F1	0.6646
ROC-AUC	0.7731
Strongest positive coefficient	Average Purchase Interval: 0.8286
Strongest negative coefficient	Median Purchase Interval: -0.4052

Deliverable

- Data/Processed/logistic_regression_predictions.csv

Business meaning

Day 5 established a measurable and interpretable benchmark against which more complex models could be compared.

3. DAY 6 — Model Comparison

Objective

Determine whether nonlinear ensemble methods provide better predictive performance than the Logistic Regression baseline.

Algorithms

- Logistic Regression — baseline/interpretable linear classifier.
- Random Forest — ensemble tree-based classifier; 100 estimators; max depth unconstrained.
- Gradient Boosting — boosting-tree classifier; 100 estimators; learning rate 0.1; max depth 3.

Verified findings

Model / finding	Verified result
Leading candidate	Gradient Boosting
Gradient Boosting ROC-AUC	0.7769
Gradient Boosting F1	0.6990
Improvement vs Day 5 ROC-AUC	+0.0038
Random Forest train ROC-AUC	1.0000
Random Forest test ROC-AUC	0.7484
Random Forest accuracy gap	30.50% — severe overfitting concern
Logistic Regression accuracy gap	-0.72% — strong stability
Gradient Boosting accuracy gap	4.49% — low overfitting concern

Deliverable

- Data/Processed/model_comparison_results.csv

Decision

Gradient Boosting became the strongest candidate for further validation. The improvement over Logistic Regression is limited, so Day 6 is model selection evidence—not proof of production superiority.

4. DAY 7 — ML Validation

Objective

Subject all candidate models to additional validation to evaluate stability, generalization, leakage controls, classification errors, probability quality, thresholds, and feature influence.

Validation design

- 3-fold StratifiedKFold cross-validation.
- Stratified 80/20 train/test split with random_state=42.
- Models: Logistic Regression, Random Forest, Gradient Boosting.
- Random Forest: 100 estimators.
- Gradient Boosting: 100 estimators.
- No hyperparameter tuning and no nested cross-validation.

Analytical sections

- Overfitting assessment.
- Confusion matrices.
- ROC curves and Precision-Recall curves.
- Probability distributions.
- Threshold analysis from 0.30 to 0.70.
- Feature importance / coefficient analysis.

Execution record

Item	Result
Execution	PASS
Best model	Gradient Boosting
ROC-AUC	0.7769
F1	0.6990
Execution time	~60 seconds
Errors	None
Source data modified	NO
Days 1–6 modified	NO

Deliverables

- Data/Processed/ml_validation_results.csv
- Docs/06_confusion_matrices.png
- Docs/07_roc_curves.png
- Docs/08_precision_recall_curves.png
- Docs/09_probability_distribution.png
- Docs/10_threshold_analysis.png

- Docs/11_feature_importance.png

5. DAY 8 — Customer Risk Scoring

Objective

Operationalize the selected Gradient Boosting approach into customer-level churn probabilities and business-oriented risk tiers.

Inference workflow

Customer behavior → ML model → churn probability → risk segment → future campaign/dashboard/AI action

Risk segmentation

Segment	Probability rule	Customers	Share
HIGH	≥ 0.70	690	18.62%
MEDIUM	$0.40 \leq P < 0.70$	1,511	40.78%
LOW	< 0.40	1,504	40.59%
TOTAL	—	3,705	100%

Verified results

- All 3,705 eligible customers received a churn probability and risk segment.
- Mean churn probability: 0.465952.
- Risk-segment constraints passed.
- One record per customer was produced.

Deliverables

- Data/Processed/customer_risk_scores.csv
- Docs/12_risk_segment_distribution.png
- Notebook/08_customer_risk_scoring.ipynb

Business caveat

The 0.70 and 0.40 cutoffs are prototype business thresholds. They should later be tested against retention economics, campaign capacity, customer value, and actual intervention outcomes.

6. End-to-End Architecture After Day 8

Layer	Role	Status
Raw data	Original UCI Online Retail II transactions	Existing
Data cleaning	Create reliable transaction dataset	Completed earlier
Customer behavior	Customer-level behavioral features	Completed earlier
Churn methodology	Define observed validation outcome	Completed earlier
ML baseline	Interpretable Logistic Regression	Day 5 complete
Model comparison	Compare three candidate algorithms	Day 6 complete
ML validation	Test stability/generalization/diagnostics	Day 7 complete
Risk scoring	Probability → HIGH/MEDIUM/LOW	Day 8 complete
SQL layer	Operational queries and business KPIs	Next phase
Power BI	Decision dashboard and monitoring	Next major phase
AI layer	Risk explanations and recommended actions	Later phase
Prototype/SaaS	Usable customer-facing product workflow	Later phase

CustomerIQ product loop: Detect → Prioritize → Explain → Act → Measure

The predictive foundation is now strong enough to move toward operational business tooling rather than continuing notebook work indefinitely.

7. Risks, Assumptions & Limitations

- **Observed churn limitation:** the target represents observed non-return during a fixed 90-day window, not a universal or causal churn definition.
- **Model performance:** ROC-AUC of 0.7769 indicates useful ranking signal but does not imply certainty.
- **Model selection:** Gradient Boosting narrowly outperformed the baseline; further business validation is required.
- **Overfitting:** Random Forest showed severe overfitting and should not be selected without substantial redesign/regularization.
- **Risk thresholds:** HIGH/MEDIUM/LOW thresholds are prototype rules, not optimized intervention policies.
- **Deployment:** no live production inference, monitoring, CRM integration, or automated retention campaign has yet been established.

Primary assumptions

- Historical transaction behavior contains predictive information about future observed non-return.
- The six behavioral predictors are sufficient for the current prototype.
- Business users can act on customer-level risk segmentation.

8. Official Completion Record

Day	Status	Primary artifact
Day 5	COMPLETE	Logistic Regression baseline + logistic_regression_predictions.csv
Day 6	COMPLETE	Model comparison + model_comparison_results.csv
Day 7	COMPLETE	ML validation + ml_validation_results.csv + six diagnostics
Day 8	COMPLETE	Customer risk scoring + customer_risk_scores.csv + segment plot

Official phase conclusion: Days 5–8 completed CustomerIQ's predictive intelligence foundation. The system now converts historical customer behavior into a validated churn-risk estimate and an actionable risk segment for 3,705 eligible customers.

Next strategic move: stop treating the notebook as the final product. Move the validated outputs into SQL and Power BI, then build the AI explanation/recommendation layer and a usable prototype.

This document records only the metrics and specifications supplied in the verified Day 5–8 project summary.